FoodWarden Proposal

Problem And Target User

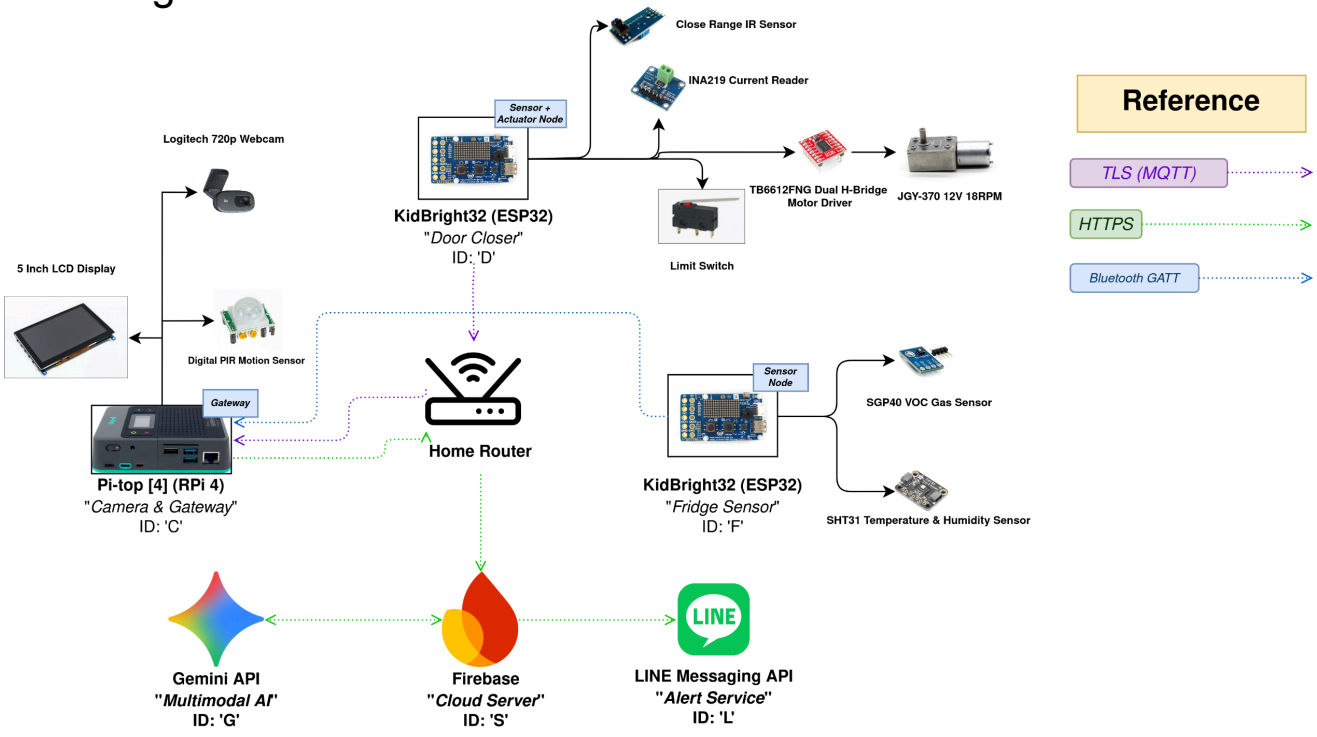
Problem: Food waste caused by perishables spoiling inside household refrigerators.
Target User: Households and consumers with shared refrigerators.

The ***FoodWarden*** system addresses a widespread household issue: food waste. It specifically targets refrigerator waste, where perishables frequently spoil within days if left unmonitored.

FoodWarden tackles this problem by resolving three preventable points of failure:

1. **Food forgotten in the back of the fridge that expires and gets thrown away.**
   We solve this with a food-logging station mounted on the front of the fridge. Users can quickly check items in and out, while keeping track of their active inventory in real time via an online dashboard.
2. **Unnoticed food spoilage that releases decay gases and bacteria, ruining surrounding fresh food.**
   We solve this with a dedicated sensor node placed inside the compartment. The sensor continuously monitors air quality (VOCs) and temperature to catch spoiling food early and prevent cross-contamination.
3. **Perishables warming up when the refrigerator door is accidentally left open.**
   We solve this with a motorized arm mounted on top of the fridge. Working alongside the front camera to detect human presence, the arm safely pushes the door shut if it is left ajar past a set time limit.

Block Diagram



Bill Of Materials

Subsystem	Component	Qty	Unit Price	Subtotal
C	5-inch HDMI TFT LCD (800×480)	1	Owned	-
C	Digital PIR Motion Sensor	1	Owned	-
C	Pi-top [4]	1	Owned	-
C	Logitech 720p Webcam	1	Owned	-
D & F	Kidbright 32	2	Owned	-
D	TCRT5000 Digital IR Obstacle Sensor	1	Owned	-
D	INA 219 Current Reader	1	119	119
D	KW11-3Z Roller Lever Micro Limit Switch	1	12	12
D	Toshiba TB6612FNG Dual H-Bridge Driver	1	59	59
D	JGY-370 12V 18RPM Worm Gear Motor	1	220	220
D	VSTARCAM 12V 2A DC Wall Adapter + Female Jack	1	75	75
D	3D Printed Mount & Sweeper Arm (ABS)	TBD	10 Baht/gram	TBD
D	LM2596 DC-DC Buck Converter	1	49	49
F	SGP40 VOC Gas Sensor	1	262	262
F	SHT31 Temperature & Humidity Sensor	1	90	90
F	18650 Battery Shield V3	1	167	167
F	Vapcell INR18650 M35 3500mAh	2	145	290
F	Sealed Food Container + Silica Gel Desiccant	1	Owned	-
S	Firebase, Gemini API, LINE API	1	0 (Free tier)	0

Total cost (excluding shipping, vouchers and 3D printed items): 1343 Baht

Embedded Engineering Depth Items

We will choose E1, E2, E3, E5, E7 as the engineering depth items that will be completed.

Member Roles

6833046321	Chawin Sinman	Embedded Developer (Gateway & Cloud) UI/UX & Dashboard Developer
6833130321	Nontapat Auetrongjit	Embedded Developer (Gateway & Cloud)
6833150921	Pranithi Tungmunkij	Embedded Developer (Gateway & Cloud) UI/UX & Dashboard Developer
6833254521	Sira Tongsima	Project Manager & Test Lead System Architect & Integration
6833288921	Alangkorn Thongkhong	Embedded Developer (Sensor Node)